EXPERIMENT-3

ROLL NO:240701221 **NAME:KAARTHIKA S**

USER INTERFACE AND DESIGN

Difference between CLI (Command line interface) and GUI (Graphical user interface) and VUI (Voice user interface)

AIM:

To design and implement a file renaming system using Command Line Interface (CLI), Graphical User Interface (GUI), and Voice User Interface (VUI), and compare their usability and efficiency.

Command Line Interface (CLI)

A Command Line Interface allows users to interact with the system by typing commands into a terminal.

Implementation:

The CLI program renames files using command-line arguments.

Working:

- Accepts old filename and new filename as inputs
- Uses os.rename() function
- Displays success or error messages

```python
import sys

def rename_file(old_name, new_name):
  try:
     os.rename(old_name, new_name)
     print(f"File renamed from {old_name} to {new_name}")
  except FileNotFoundError:
     print(f"Error: {old_name} not found.")
  except Exception as e:
     print(f"An error occurred: {e}")

if __name__ == "__main__":
  if len(sys.argv) != 3:
     print("Usage: python rename_file_cli.py <old_filename> <new_filename>")
  else:
     rename_file(sys.argv[1], sys.argv[2])
```

Output:

PS D:\exp3> python cli.py old.txt new.txt
File renamed from old.txt to new.txt

Graphical User Interface (GUI)

A GUI allows users to interact through graphical elements such as windows, buttons, and text fields.

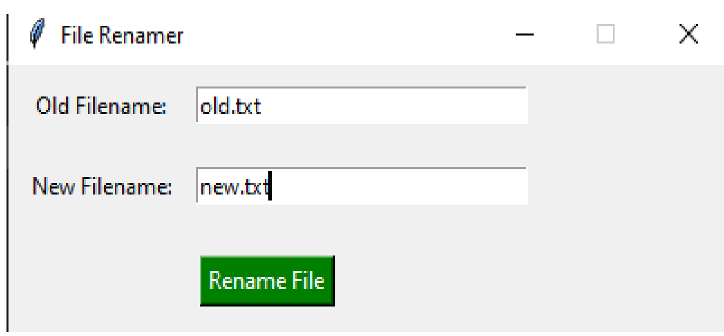
Implementation:

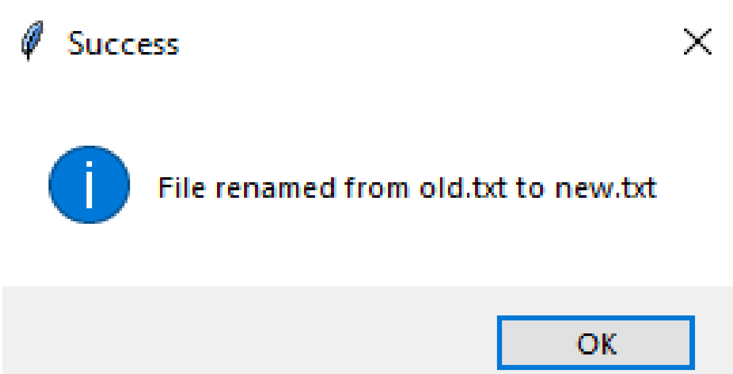
The GUI application is built using Tkinter, allowing users to rename files through a graphical window.

```python
Import tkinter as tk
from tkinter import messagebox
import os

def rename_file():
    old_name = old_filename_entry.get()
    new_name = new_filename_entry.get()

    if not old_name or not new_name:
        messagebox.showwarning("Warning", "Please enter both filenames.")
        return

    try:
        os.rename(old_name, new_name)
        messagebox.showinfo("Success", f"File renamed from {old_name} to
{new_name}")
    except FileNotFoundError:
        messagebox.showerror("Error", f"File '{old_name}' not found.")
    except Exception as e:
        messagebox.showerror("Error", f"An error occurred: {e}")

root = tk.Tk()
root.title("File Renamer")
root.geometry("400x150")    # makes window look professional
root.resizable(False, False)

tk.Label(root, text="Old Filename:").grid(row=0, column=0, padx=10, pady=10)
tk.Label(root, text="New Filename:").grid(row=1, column=0, padx=10, pady=10)
```

```
old_filename_entry = tk.Entry(root, width=30)
old_filename_entry.grid(row=0, column=1)

new_filename_entry = tk.Entry(root, width=30)
new_filename_entry.grid(row=1, column=1)

rename_button = tk.Button(root, text="Rename File", bg="green", fg="white",
command=rename_file)
rename_button.grid(row=2, columnspan=2, pady=15)

root.mainloop()
```

Output: PS D:\exp3> python gui.py





Voice User Interface (VUI)

A Voice User Interface enables interaction using voice commands and speech recognition technology.

Implementation:

The VUI system uses speech recognition to rename files based on voice commands.The spoken command is converted into text, and the file is renamed using Python

```
import speech_recognition as sr
import os
recognizer = sr.Recognizer()
try:
  with sr.Microphone() as source:
      print("Say: rename old.txt to new.txt")
      recognizer.adjust_for_ambient_noise(source)
      audio = recognizer.listen(source)

  try:
      command = recognizer.recognize_google(audio)
      print("You said:", command)
```

```python
        words = command.lower().split()
        corrections = {
            "text": "txt",
            "pdf": "pdf",
            "doc": "doc"
        }
        reconstructed = []
        i = 0
        while i < len(words):
            if i + 2 < len(words) and words[i + 1] == "dot":
                # Merge word + dot + extension
                extension = words[i + 2]
                # Apply corrections to extension
                if extension in corrections:
                    extension = corrections[extension]
                reconstructed.append(words[i] + "." + extension)
                i += 3
            else:
                reconstructed.append(words[i])
                i += 1
        old_name = reconstructed[1]
        new_name = reconstructed[3]

        os.rename(old_name, new_name)
        print("File renamed successfully!")

    except sr.UnknownValueError:
        print("Error: Could not understand audio")
    except sr.RequestError as e:
        print(f"Error: API request failed - {e}")
    except Exception as e:
        print("Error:", e)

except OSError as e:
    print(f"Error: Microphone not available - {e}")
```

```
print("Please ensure a microphone is connected and properly configured.")
```

.**Output:**

PS D:\exp3>.\.venv311\Scripts\Activate.ps1
>>
(.venv311) PS D:\exp3> python vui.py
Say: rename old.txt to new.txt
You said: rename old dot text to new dot text
File renamed successfully!

RESULT:

The file renaming system was successfully implemented using CLI, GUI, and VUI. Each interface allowed users to rename files effectively while demonstrating different levels of usability.